

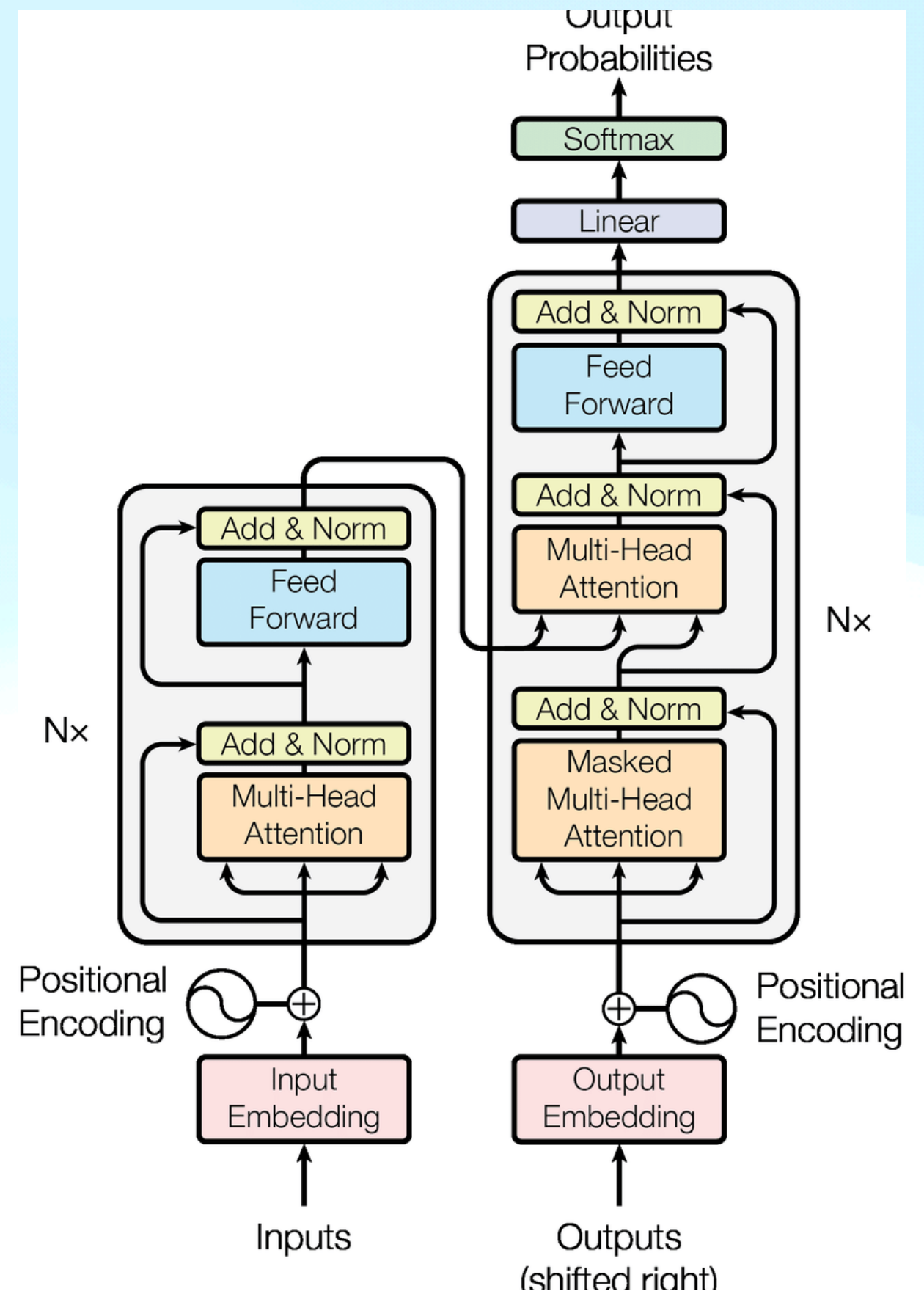
Numpy Transformer

Goal

- Implement a transformer only using numpy
- Teach the model to do addition

Architecture

- Embedding
- Positional Encoding
- Encoder
- Decoder



Embedding and Positional Encoding

- Embedding: converts token IDs $\rightarrow$ vectors
- Positional encoding: adds position information directly to embeddings

$$PE_{(pos,2i)} = \sin(pos/10000^{2i/d_{model}})$$

- $PE_{(pos,2i+1)} = \cos(pos/10000^{2i/d_{model}})$

Attention

- Compute Q(Query), K(Key), V(Value) with linear layers
- Split into heads

- Compute $\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V$

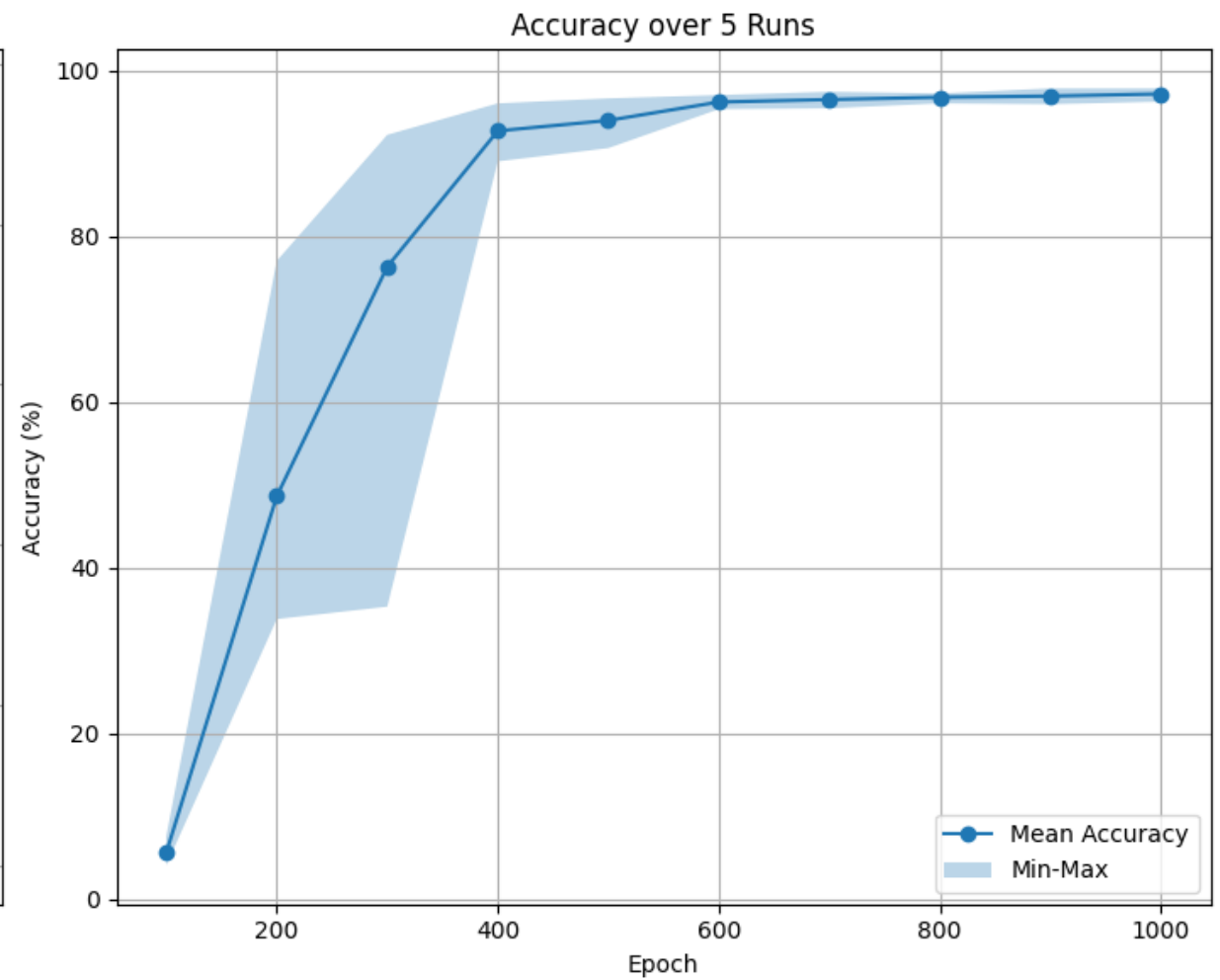
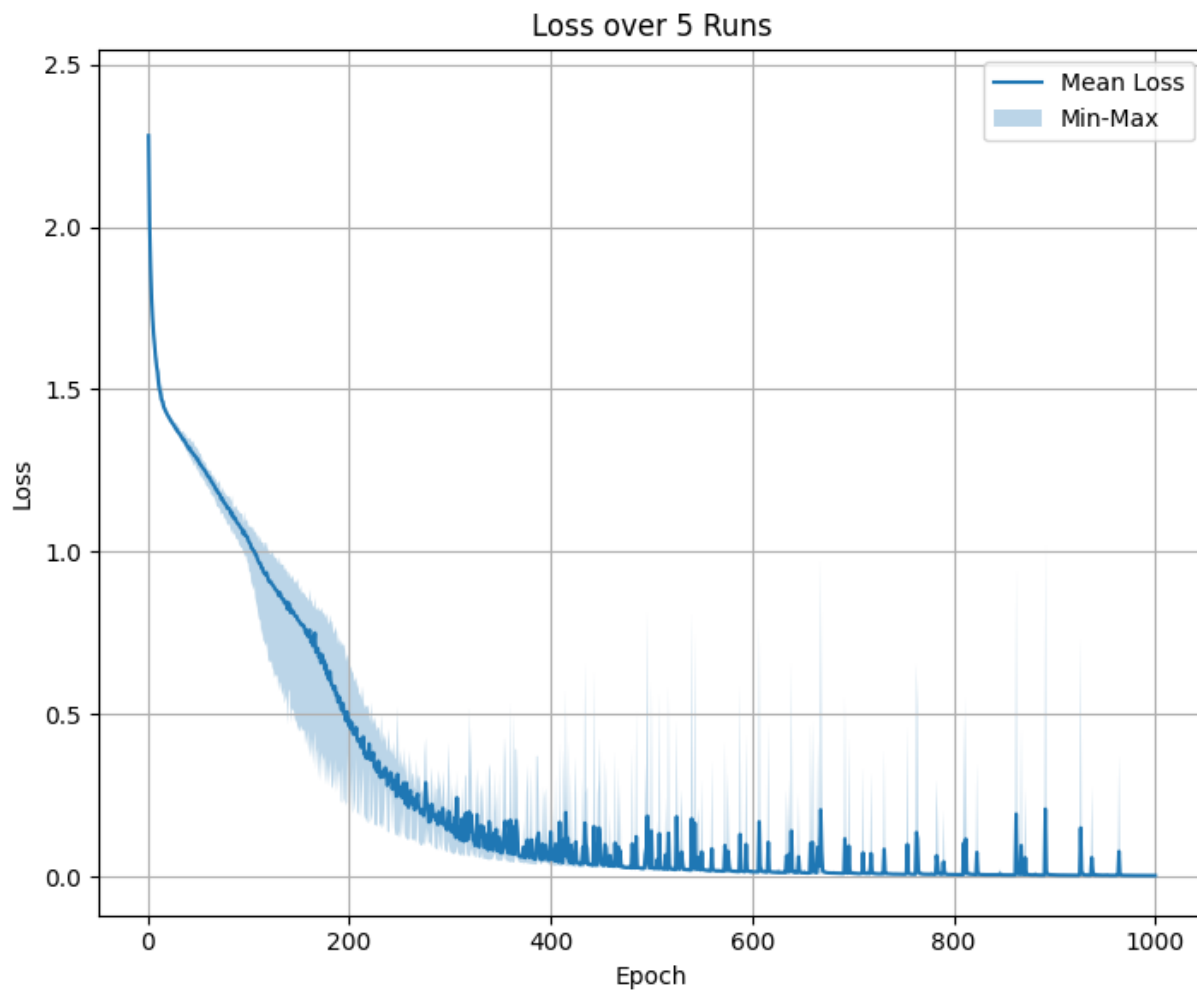
Encoder and Decoder

- Encoder
 - Self-attention
 - Feed-forward
 - LayerNorm + residuals
- Decoder
 - Masked self-attention
 - Cross-attention with encoder output
 - Feed-forward
 - LayerNorm + residuals

Training

- Two-digit addition
- Dataset: all possible combinations of two-digit
- Train/test split: 80/20
- Loss: cross-entropy
- Optimizer: SGD
- Epochs: 1000×5 runs

Results



Example

- Mostly correct
- Made a mistake at $7+0$
- Some other mistakes it made:
 - $44+45=99$
 - $3+7=11$

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[Run 5 | Epoch 1000] EXAMPLES
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```

```
59+28 = 87 | Predicted: 87
95+44 = 139 | Predicted: 139
63+3 = 66 | Predicted: 66
24+61 = 85 | Predicted: 85
29+60 = 89 | Predicted: 89
14+98 = 112 | Predicted: 112
66+55 = 121 | Predicted: 121
9+70 = 79 | Predicted: 79
63+82 = 145 | Predicted: 145
29+29 = 58 | Predicted: 58
7+0 = 7 | Predicted: 3
29+22 = 51 | Predicted: 51
4+84 = 88 | Predicted: 88
32+81 = 113 | Predicted: 113
41+89 = 130 | Predicted: 130
82+22 = 104 | Predicted: 104
43+96 = 139 | Predicted: 139
82+80 = 162 | Predicted: 162
7+83 = 90 | Predicted: 90
26+87 = 113 | Predicted: 113
=====
```

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[Run 5 | Epoch 1000] Accuracy = 97.20%
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```

Why Mistakes Happen

- Small dataset
- Carrying across digits is harder
- Model is small ($d_{\text{model}}=32$)
- Training uses SGD
- No regularization

Conclusion

- NumPy Transformer works well
- Learns two-digit addition
- High accuracy but small mistakes made

Thank You!